

Lab Manual — Week 8

Concurrency & Distributed Systems — Lebanese University, Faculty of Engineering

Topic: Midterm Debugging and Kafka Mental-Model Preview

Learning Objectives

By the end of Week 8, you will be able to:

1. Interpret a thread-dump clue.
2. Write a mechanism-first diagnosis report.
3. Fix an async request that can wait forever.
4. Explain the performance tradeoff of timeout/fallback behavior.
5. Deliver a five-minute technical debugging explanation.
6. Explain Kafka's log model as history, replay, recovery, audit, and decoupling.

Lab Folder

```
week8-lab/week8-day1-midterm-debugging-kafka-preview
```

Use:

```
.\gradlew.bat test
.\gradlew.bat studentCheck
.\run-all-demos.ps1
```

Required Midterm Submission

- Thread dump evidence or simulated thread-dump interpretation.
- Diagnosis report.
- Fixed code.
- Performance tradeoff analysis.
- Five-minute recorded explanation.
- Kafka preview answer.

Part 1 — Diagnosis Report

Open:

```
MIDTERM_PACKET.md
part1_diagnosis/exercises/Ex1_DiagnosisReport.java
```

Task: Identify the bug, evidence, mechanism, and fix.

Rule: Do not stop at "it hangs." Explain why it hangs.

Part 2 — Async Server Fix

Open:

```
part2_async_server/exercises/Ex2_FixAsyncServer.java
```

Task: Protect a never-completing dependency with timeout fallback.

Expected behavior:

```
fallback-response
```

Part 3 — Performance Tradeoff

Open:

```
part3_tradeoffs/exercises/Ex3_PerformanceTradeoff.java
PERFORMANCE_TRADEOFF.md
```

Task: Explain benefit, cost, and measurement plan.

Field	Expected content
Benefit	Bounded waiting protects request threads.
Cost	Slow real responses may become fallback responses.
Measurement	Latency percentiles, timeout count, fallback rate.

Part 4 — Kafka Preview

Open:

```
KAFKA_PREVIEW.md
part4_kafka_preview/exercises/Ex4_KafkaMentalModel.java
```

Task: Explain log as:

- history,
- replay,
- recovery,
- audit,
- decoupling.

Five-Minute Explanation

Use `RECORDED_EXPLANATION_RUBRIC.md`. The explanation should cover:

1. Bug.
2. Evidence.
3. Mechanism.
4. Fix.
5. Cost.

Bridge to Week 9

Week 9 moves to Python and language/runtime limits. Keep the Week 8 habit: separate symptoms from mechanism, and measure before making performance claims.